

VEDANT SINGH

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EDUCATION

Bachelor of Technology in Computer Science and Engineering

Expected May 2028

VIT Bhopal University, Bhopal, India

TECHNICAL SKILLS

Languages: Java, C++, Python, JavaScript, TypeScript, SQL

Web: HTML5, CSS3, React.js, Next.js, Tailwind CSS

AI / ML: RAG, LLMs, Embeddings, Semantic Search, Prompt Engineering

Libraries & APIs: PDF.js, Marked.js, HTML5 Audio, Web Audio API

Tools / Platforms: Git, GitHub, Vercel, Android Studio, OpenRouter

EXPERIENCE

Bharat Heavy Electricals Limited (BHEL)

May 2026 – Jun 2026

Engineering Intern

Bhopal, India(On-site)

- Completed a 30-day engineering internship and gained exposure to engineering workflows, documentation, and organizational processes.
- Collaborated with professionals in an industrial environment and observed project execution and technical operations.

PROJECTS

Radio Ishq – 3D Music Player & Karaoke Experience [Live Demo](#)

Next.js, React, TypeScript, Tailwind CSS, Web Audio API

- Engineered a 3D Coverflow music player using CSS 3D transforms with drag, swipe, wheel gestures, keyboard navigation, and animated track transitions across an 11-track playlist.
- Developed an HTML5 audio engine supporting seeking, volume control, shuffle, repeat modes, queue management, continuous playback, and live frequency-bar visualization.
- Implemented synchronized karaoke lyrics using real-time audio events, active-line detection, and automatic lyric scrolling.
- Built a dynamic visual atmosphere that adapts gradients to the active track and applied responsive glassmorphism-based controls.

StudyMate AI – AI-Powered Academic Assistant [Live Demo](#)

JavaScript, PDF.js, OpenRouter, RAG, Embeddings

- Developed a browser-based academic assistant combining AI chat, PDF study notes, quiz generation, personalized study planning, and progress tracking.
- Implemented a Retrieval-Augmented Generation pipeline that extracts PDF text, creates overlapping chunks, generates embeddings, and retrieves relevant context before answer generation.
- Combined semantic similarity with keyword relevance for hybrid retrieval and preserved page metadata to surface source references.
- Integrated AI workflows for academic Q&A, five-question MCQ generation with scoring and retry, and day-by-day study plans.

LEADERSHIP & ADDITIONAL

- Maintained public GitHub repositories and documented end-to-end software and AI projects with architecture, implementation, and setup details.